

Technical Proposal: [System Architecture Name: TBD]

Project Name: [TBD]

1. System Configuration

This architecture separates the optical interface from the primary computational and sensor-processing hardware to achieve an ultra-lightweight form factor.

- **Head-Worn Unit [Name: TBD]:** A pair of spectacles designed to be indistinguishable from standard everyday glasses. The target weight is **30g**.
- **Host Unit [Name: TBD]:** A headless laptop chassis containing a high-performance SOC, a dedicated FPGA for sensor processing, a battery, and physical input devices (keyboard and trackpad).
- **Interconnect [Name: TBD]:** A unified cable providing power and high-bandwidth data transmission between the host unit and the spectacles.

2. Optical System and Field of View

The system aims for a **125° diagonal Field of View (FoV)** using the following optical logic:

- **Optics [Tech Name: TBD]:** Fisheye lenses are employed to provide a broad physical field of vision.
- **Selective Distortion Correction:** Digital inverse distortion is applied to the central region of the view for geometric accuracy.
- **Threshold-Based Cropping:** Peripheral regions where lens distortion exceeds a specific quality threshold are not computationally corrected. Instead, these areas are **cropped**. This ensures a clean 125° diagonal view focused on high-fidelity areas.

3. Rendering Logic (Anstis 1974)

The system utilizes a foveated rendering method [Tech Name: TBD] based on the visual acuity research of **Anstis (1974)**.

- **Gaze-Tracked Resolution:** Only the foveal center, identified by high-speed eye-tracking, is rendered at **60 PPD (Pixels Per Degree)**.
- **Peripheral Gradation:** Rendering resolution is stepped down progressively towards the edges of the field of view, following the natural decline of human retinal sensitivity.

4. Externalized Sensor Processing

To minimize weight and heat on the head-worn unit, all significant processing is moved to the host unit.

- **Host-Side FPGA [Module Name: TBD]:** Sensor data—including eye-tracking, IMU, and 6DoF tracking—is transmitted from the spectacles and processed by a **dedicated FPGA located within the host laptop chassis**.
- **Latency Management:** The FPGA handles sensor fusion and image reprojection to maintain visual stability and minimize Motion-to-Photon latency.

5. Interaction and Workspace Definition

- **Workspace Instantiation [Method Name: TBD]:** Virtual monitors are created via 3D spatial gestures. A user instantiates a virtual display by **drawing a rectangle in 3D space with their fingertips**.
- **Spatial Mapping:** The system recognizes the coordinates of the drawn rectangle and anchors a virtual screen at that specific location.
- **Primary Input:** Professional tasks are managed through the physical keyboard and trackpad of the host unit, with eye-gaze used for window focus and navigation.

6. Design Objective

The objective is to provide a high-performance spatial workstation within the ergonomics of traditional eyewear. By removing the display from the host laptop, the system redirects the associated power and thermal budget to the SOC and FPGA for stable processing.